

# **AI for Economic Research: Dynamic Models, Language, and Agents**

## **Lecture 10.5: Case Study 4 — AI as Research Collaborator**

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Spring 2026

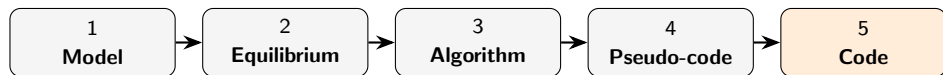
## Topic 10.5 Agenda

1. **Reference framework** — the canonical five-stage research workflow
2. **Case Study 4** — AI as research collaborator across the lifecycle

## **Reference framework**

Model → equilibrium → algorithm → pseudo-code → code

# The Canonical Five-Stage Workflow



- **Stages 1–4** are *contracts*. They stop the agent from inventing the problem while coding.
- **Stage 5** is execution. It is allowed only after the contracts are clear.
- **Human approval** happens between stages. Skipping a stage is a process failure, even if the code runs.

## Each Stage Produces a Reviewable Artifact

| Stage          | Artifact                                                                  | Exit gate (human-checked)                                  |
|----------------|---------------------------------------------------------------------------|------------------------------------------------------------|
| 1. Model       | <code>delta_spec.md</code> : equations, parameters, what stays unchanged  | every new symbol has a defining condition                  |
| 2. Equilibrium | policy, prices, distributions, aggregates, and clearing conditions        | every new market has a clearing equation                   |
| 3. Algorithm   | method choice + numerical knobs + diagnostics                             | where it can fail; what convergence criterion              |
| 4. Pseudo-code | <code>pseudocode.md</code> : loop nesting, tensor shapes, validation gate | a 2nd-year PhD could implement without follow-up questions |
| 5. Code        | source files, tests, session notes                                        | validation gate passes; reduce-to-previous behaves         |

**Mantra.** Stages 1–4 are *contracts*; Stage 5 is *execution*. The researcher's job is to design and approve the contracts.

# Prompt Patterns for Stages 1–4

```
# Stage 1 (Model) prompt pattern
Read versions/v{n}/model_spec.md.
Draft versions/v{n+1}/delta_spec.md for exactly one model extension.
Define equations, parameters, outputs, and what stays unchanged.
Restate the model in plain language; wait for my approval.

# Stage 2 (Equilibrium) prompt pattern
State the recursive equilibrium. List the policy, prices, distributions,
aggregates, and all clearing/optimality conditions. Push back on any
inconsistency in my model spec.

# Stage 3 (Algorithm) prompt pattern
Compare candidate methods and trade-offs. I am choosing {method}
because {reason}. Explain where it may fail, what diagnostics to log,
and what convergence criterion to use. Do not write code.

# Stage 4 (Pseudo-code) prompt pattern
Write Python-style pseudo-code: loop nesting, tensor shapes, loss terms,
diagnostics, and validation gate. Then have domain-reviewer check
Euler/KKT/market-clearing equations.
```

## Stage 5 (Code): Implementation and Validation

**Artifact.** Source files, tests, notebook section, and `session_notes.md`.

```
Implement  $V_{n+1}$  per the approved pseudocode. Include diagnostics,  
tests/test_v_{n+1}_smoke.py, and the validation gate in delta_spec.md.  
Do NOT add features beyond  $V_{n+1}$ . Plan first; show me the plan  
before writing code. Then run the tests and iterate until the gate  
passes.
```

**Agent loop.** Plan  $\rightarrow$  approve  $\rightarrow$  implement  $\rightarrow$  verify  $\rightarrow$  review  $\rightarrow$  fix  $\rightarrow$  score.

**Stop condition.** The validation gate passes, *and* reduce-to-previous checks behave as expected (e.g.,  $\psi_K = 0$  recovers the prior version).

**Key lesson.** A passing test suite is not enough; the gates must encode *economics* (procyclical capital, lifecycle hump, market clearing).

## Mapping the Five Stages onto Case Study 4 (Labor Market)

| Canonical stage | Case 4 instantiation                                            | Artifact                           |
|-----------------|-----------------------------------------------------------------|------------------------------------|
| 1. Model        | glue-work theory; AI-Easy vs. AI-Hard glue; $2 \times 2$ matrix | verbal theory + O*NET item codes   |
| 2. Equilibrium  | empirical specification; allowed predictions P1, P2             | specification prompt + claims memo |
| 3. Algorithm    | 11-script pipeline architecture; <code>config.py</code>         | pipeline DAG + 51 unit tests       |
| 4. Pseudo-code  | per-script validation rules + adversarial review                | module table + review memo         |
| 5. Code         | scripts run; data executed; results interpreted                 | figures + repositioned paper       |

**Same vocabulary, different domain.** Stages 1–4 remain contracts; Stage 5 is execution. The discipline travels from dynamic macro solvers to empirical labor-market research *intact*.

## **Case Study 4**

AI as full-lifecycle research collaborator in empirical macroeconomics

## Case 4 — The Research Question

**Question:** Why has rapid AI adoption not produced uniform occupational collapse, and what role does coordinative labor play in shaping which occupations survive?

- Paper: “Glue Work, Task Bundles, and the AI Labor Market” — Feng (2026, in progress)
- Extends Garicano–Li–Wu (2026) by endogenizing bundle strength via glue labor supply
- **This case covers the full research lifecycle:** theory articulation, empirical pipeline design, adversarial review, and claims discipline — not just one execution stage

# The Glue Work Hypothesis

## What is glue work?

- Coordinative tasks that integrate components without producing a visible standalone output
- O\*NET examples: coordinating work and activities; scheduling; communicating across silos
- Underrewarded: workers capture only  $\lambda < 1$  of the social marginal product
- When glue erodes, the bundle weakens — often with a lag before it becomes visible

## The 2×2 AI impact matrix

|               | Low glue    | AI | High glue       | AI |
|---------------|-------------|----|-----------------|----|
| High AI prod. | Restructure |    | Full disruption |    |
| Low AI prod.  | Stable      |    | Glue erosion    |    |

Core prediction: high-AI + high-glue occupations *restructure* rather than collapse.

**The insight was human. The matrix formalization was joint.**

## Stage 1 — Theory Building with AI

- Human developed the core insight: endogenize bundle strength through glue labor supply
- Claude was asked to:
  - Articulate the  $2 \times 2$  prediction matrix from verbal intuition
  - Define AI-Easy Glue vs. AI-Hard Glue using specific O\*NET item codes
  - Identify the market failure formally ( $\lambda < 1$  creates underinvestment equilibrium)
- **AI-Easy Glue** (LLM-susceptible): documenting/recording information, synthesizing knowledge, routine cross-functional reporting
- **AI-Hard Glue** (embodied/relational): building trust, mentoring, navigating ambiguous authority, reading implicit organizational signals
- Prompting mode: **specification-driven** — “List O\*NET items 4C1–4C3 that qualify as AI-Hard Glue by the following criteria...” not “tell me what glue work is”

## Stage 2 — Translating Theory into Empirical Specification

```
## Empirical Specification Prompt
```

```
You are translating the following theoretical model into a  
testable empirical specification.
```

```
THEORY CLAIMS:
```

- P1: High-glue, high-AI-exposure occupations restructure rather than collapse (employment share positive 2019-2024)
- P2: The reallocation margin operates at junior entry (age < 30), not incumbent exit

```
VARIABLE CONSTRUCTION RULES:
```

- glue\_core: mean of O\*NET items [4C1.a.2, 4A4.a.1, 4A4.b.1] normalized to [0,1] at SOC 2018 level
- ai\_exposure: FRS AIOE score (Felten-Raj-Seamans 2021)
- outcome: CPS monthly employment share (seasonally adj.)

```
FORBIDDEN: do not use occupation-level wage as a glue proxy  
without first checking correlation > 0.6 with glue_core.
```

- Exact item codes prevent hallucinated variable definitions
- A “FORBIDDEN” instruction enforces scope before results appear

# Claims Discipline — Setting the Rules Before Seeing Results

```
## Claims Discipline Specification

BEFORE any data is analyzed, record the following:

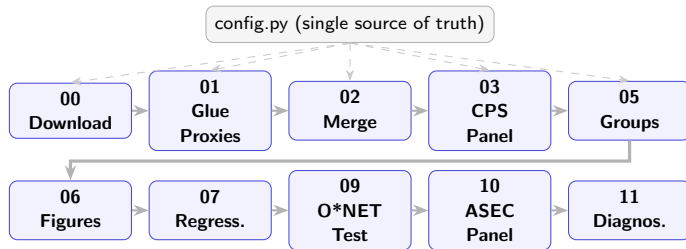
ALLOWED CLAIMS (theory-derived, testable):
- P1: Occupations in top quartile of both glue_core and AIOE
  show positive employment share growth 2019--2024
- P2: Entry share (age < 30) declines faster in
  low-glue / high-AI occupations

FORBIDDEN CLAIMS (data-mining temptations to block):
- Any claim about wage effects not pre-specified here
- Any interaction term not listed in the model section
- Any subgroup finding discovered post-estimation

GATE: results memo must cite which allowed claim each
regression speaks to before interpretation proceeds.
```

**This is claims discipline.** It must be set before results appear, not reconstructed after. It is the empirical analog of pre-registration.

## Stage 3 — The 11-Script Pipeline Architecture



**One config.py = single source of truth.** 51 unit tests specified before any script runs on real data.

## Pipeline Module Detail

| Script | Key output                                  | Validation rule                                     | Each |
|--------|---------------------------------------------|-----------------------------------------------------|------|
| 00     | onet_raw, frs_aioe, crosswalks (parquet)    | row count $\geq$ expected; no null SOC codes        |      |
| 01     | glue_proxies (6 variants, SOC 2018)         | all variants in $[0, 1]$ ; correlation matrix saved |      |
| 02     | master_soc: glue + AI exposure + crosswalks | merge rate $\geq 95\%$ ; no duplicate SOC           |      |
| 03     | cps_panel: employment share, entry share    | weights sum to civilian pop; seasonal adj. flag set |      |
| 05     | analysis: 4 AI $\times$ glue groups defined | group counts balanced; no SOC in $> 1$ group        |      |
| 06–11  | figures, tables, diagnostics                | files written; no NaN in reported statistics        |      |

script has explicit inputs (parquet), explicit outputs (parquet), and a pass/fail validation rule. Failure halts the pipeline.

## Stage 3 — Variable Construction Details

### Glue proxies (6 variants)

- glue\_core: 3 O\*NET items, baseline
- glue\_ext: 8-item extended set
- glue\_aihard: AI-Hard items only
- glue\_social: social/interpersonal subset
- glue\_dominance: mean glue / mean all activities
- glue\_minus\_mgmt: robustness check

The 6 variants test whether the theory's protection story depends on which aspect of coordination is measured.

### AI exposure and outcomes

- AI exposure: FRS AIOE (2021), SOC-linked
- Outcomes: CPS monthly employment share
- Entry margin: age < 30 share within occ-month
- Long-run: ASEC panel 2000–2025
- 4 groups: median splits of AIOE  $\times$  glue\_core

### Core group definitions:

|                     |               |
|---------------------|---------------|
| High AI + High Glue | Restructuring |
| High AI + Low Glue  | Disruption    |
| Low AI + High Glue  | Insulated     |
| Low AI + Low Glue   | Stable        |

## Stage 4 — Adversarial Review

**Adversarial prompt:** “Find 6 structural weaknesses in this empirical design. For each, state whether it is fatal or manageable, and why.”

Six weaknesses identified:

1. AIOE is cross-sectional — does not capture within-occupation AI adoption dynamics
2. Glue proxies are task-level O\*NET ratings, not worker-level observations
3. Selection: high-glue workers may self-select into high-AI firms for unobserved reasons
4. Crosswalk from SOC 2018 to Census codes loses  $\sim 12\%$  of occupations (a validation rule catches this)
5. CPS monthly sample sizes for narrow occupations produce large standard errors
6. **The interaction ( $\text{AI} \times \text{Glue}$ ) is negative in the data** — but the theory predicts positive

**Item 6 was the decisive finding.** It repositioned the entire paper.

# When Data Contradicts Theory

**Data finding:** the estimated interaction (AI exposure  $\times$  glue score) is *negative*.

**Theory predicted:** positive — glue preserves bundles, so high-glue/high-AI should survive.

**This is not a bug.** It is information.

## What the theory predicted

- High-AI + High-Glue  $\rightarrow$  restructuring and survival
- Glue preserves value when production tasks are automated
- Expected positive interaction

## The reinterpretation

- Negative sign: high-glue workers in high-AI occupations are displaced, not preserved
- Effect operates through *occupational reallocation*, not within-occupation survival
- Theory-specific prediction  $\neq$  generic heterogeneity claim

**Claude distinguished theory-specific predictions from generic heterogeneity findings.** That distinction determines what the paper can claim.

## Division of Labor Across the Full Lifecycle

| Task                   | Human Researcher                     | AI (Claude)                                            |
|------------------------|--------------------------------------|--------------------------------------------------------|
| Core insight           | Endogenizes bundle strength via glue |                                                        |
| Literature positioning | Sets the GLW extension               |                                                        |
| Variable specs         | Co-designed, approved                | Elaborated O*NET codes; caught crosswalk inconsistency |
| Pipeline architecture  | Approved the design                  | Designed 11-script layout from scratch                 |
| Claims discipline      | Enforces the gate                    | Drafted allowed/forbidden claims                       |
| Adversarial review     | Decided which weaknesses were fatal  | Structured the 6 adversarial findings                  |
| Data execution         | Runs the scripts                     |                                                        |
| Interpretation         | Final call on repositioning          |                                                        |
| Paper writing          | Authors                              | Assists with exposition                                |

## Key Prompting Strategies in Case 4

1. **Specification-driven:** exact O\*NET item codes, not vague requests — prevents hallucinated variable definitions
2. **Constraint-based (claims discipline):** allowed and forbidden claims specified before results appear — prevents data mining
3. **Adversarial:** “find 6 structural weaknesses” not “review this memo” — prevents sycophantic validation
4. **Modular decomposition:** each script as an independent testable unit — prevents silent errors accumulating across stages
5. **Deferred execution:** pipeline fully specified before any real data is analyzed — separates design from execution, preserving pre-specification discipline

**Each strategy is a structural response to a specific failure mode, not a stylistic preference.**

## Failure Modes Specific to Case 4

1. **AI agrees too readily:** asking “does this design look good?” produces validation; adversarial mode must be explicit
2. **Vague theory → untestable claims:** “glue work matters” is not a prediction; specification-driven prompting forces operationalization
3. **Data mining disguised as exploration:** claims discipline must be logged before results; retroactive framing is indistinguishable from pre-specification to a reader
4. **Silent crosswalk errors:** the 12% SOC loss is found only because a validation rule checks merge rate; without validation rules, the pipeline produces credible-looking numbers
5. **Mechanical overclaiming:** when data contradicts theory, the temptation is ad hoc reframing; adversarial review forces the repositioning to be deliberate and documented

# The Meta-Irony: AI Doing Glue Work

## What the paper studies

- Glue work: coordinative labor that binds complex task bundles
- Holds specifications together across stages
- Tracks implicit commitments across time
- Underrewarded because its contribution is diffuse in output attribution

## What Claude did in the research itself

- Maintained specification coherence across 11 scripts and 4 research stages
- Tracked which claims were allowed before results arrived
- Structured the adversarial review to produce actionable findings
- Held the interface between theory and empirical design

**The AI is demonstrating its own research subject.**

That is not an accident — it is evidence about where AI comparative advantage lies in complex intellectual work.

## Key Lessons from Case Study 4

- AI can operate across the **full research lifecycle** — theory articulation, pipeline design, adversarial review — not just execution
- The human researcher must own the core insight, literature positioning, data execution, and final interpretation
- **Specification-driven and adversarial prompting** are structural defenses against the most common AI failure modes in research settings
- **Claims discipline** must be established before results appear; it cannot be reconstructed credibly after
- When data contradicts theory, the right move is deliberate repositioning, not ad hoc reframing; AI can structure that repositioning but cannot decide whether it is intellectually honest
- This is a new pattern: **AI as full-lifecycle research collaborator**, distinct from skill, empirical pipeline, or dataset harness